

BANK FRAUD RISK ANALYSIS DASHBOARD

Credit Score & Fraud Detection

Power BI Dashboard — Project Documentation

Project Title	Bank Fraud Risk Analysis Dashboard
Developer Role	Power BI Developer / Data Analyst
Dataset	pb_bank_fraud_detection_project-dataset.csv
Total Records	40,959 customer records
Dashboard Pages	6 pages — Dashboard sights Overview Credit Analysis Risk & Debt Customer Behavior Payment Behavior Analysis
Tools Used	Power BI Desktop, Power Query, DAX
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Date	7 August 2026

Step 1 — Objective

1.1 Dashboard Purpose

This Power BI dashboard has been developed to enable data-driven credit risk assessment and fraud detection. It transforms raw customer financial data into clear, interactive visual insights that support the bank's underwriting, compliance, and fraud prevention teams.

Primary Objectives:

- Identify and segment customers by credit quality — Good, Standard, and Poor
- Detect high-risk behavioural patterns associated with payment default and fraud
- Analyse the relationship between income, debt, interest rate, and credit score
- Provide occupation-wise and age-wise credit risk profiling
- Enable business stakeholders to make faster, evidence-based lending decisions

1.2 Target Audience

Stakeholder	How They Use This Dashboard
Credit Risk Team	Monitor portfolio credit quality, identify high-risk segments
Fraud Analysts	Flag customers with high delay days, low income, and high debt
Loan Officers	Evaluate applicant profiles before approving loans
Senior Management	Track KPIs — overall portfolio health and default risk
Compliance Department	Ensure fair lending practices across income and occupation groups

Step 2 — Dataset Description

2.1 Dataset Overview

Dataset Summary	
File Name	pb_bank_fraud_detection_project-dataset.csv
Total Records	40,959 customers
Total Columns	28 attributes per customer
Target Variable	Credit_Score (Good / Standard / Poor)
Time Period	Multi-year customer financial history
Source	PB Internal dataset

2.2 Key Columns Used

Column Name	Data Type	Description	Used In
Credit_Score	Text	Target — Good / Standard / Poor	All pages
Annual_Income	Decimal	Yearly income in INR	Overview, Credit
Outstanding_Debt	Decimal	Total current debt in INR	Risk & Debt
Interest_Rate	Integer	Annual interest rate (%)	Risk & Debt
Delay_from_due_date	Integer	Days delayed past due date	Risk & Debt
Occupation	Text	Customer's profession	Credit Analysis
Age	Integer	Customer age in years	Credit Analysis
Payment_Behavior	Text	Spend + payment value category	Behavior
Payment_of_Min_Amount	Text	Yes / No — minimum payment made	Overview
Credit_Mix	Text	Good / Standard / Bad portfolio mix	Overview
Credit_Utilization_Ratio	Decimal	% of available credit used	Behavior
Total_EMI_per_month	Decimal	Monthly EMI obligation (INR)	Risk & Debt
Monthly_Balance	Decimal	Average monthly account balance	Behavior
Num_of_Delayed_Payment	Integer	Count of delayed payments	Risk & Debt

2.3 Data Cleaning Steps (Power Query)

Bank Fraud Risk Analysis Dashboard

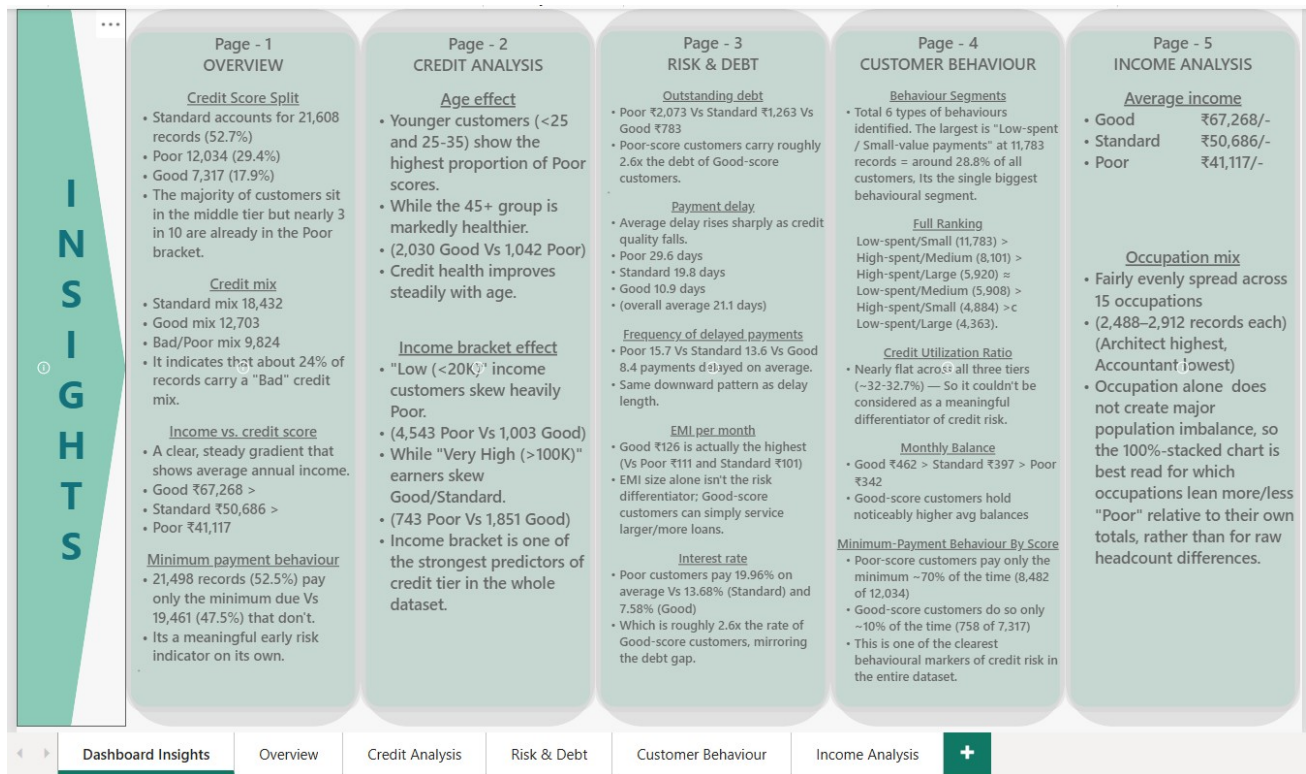
- Set Age, Num_Bank_Accounts, Num_Credit_Card, Num_of_Loan, Delay_from_due_date → Whole Number
- Set Annual_Income, Outstanding_Debt, Monthly_Balance, Total_EMI_per_month → Decimal Number
- Set Credit_Utilization_Ratio, Interest_Rate → Decimal Number
- Removed errors and nulls from Credit_Score column — only Good/Standard/Poor retained
- Removed leading/trailing whitespace from Occupation and Payment_Behavior columns
- Created 2 calculated columns: Age Group and Income Bracket using DAX SWITCH()
- Created sort-order columns: Credit Score Order and Age Group Order for correct chart sequencing

Step 3 — Dashboard Pages (Screenshot Reference)

The dashboard contains 6 navigation pages. Each page is described below with layout details. Insert actual screenshots from Power BI Desktop after publishing.

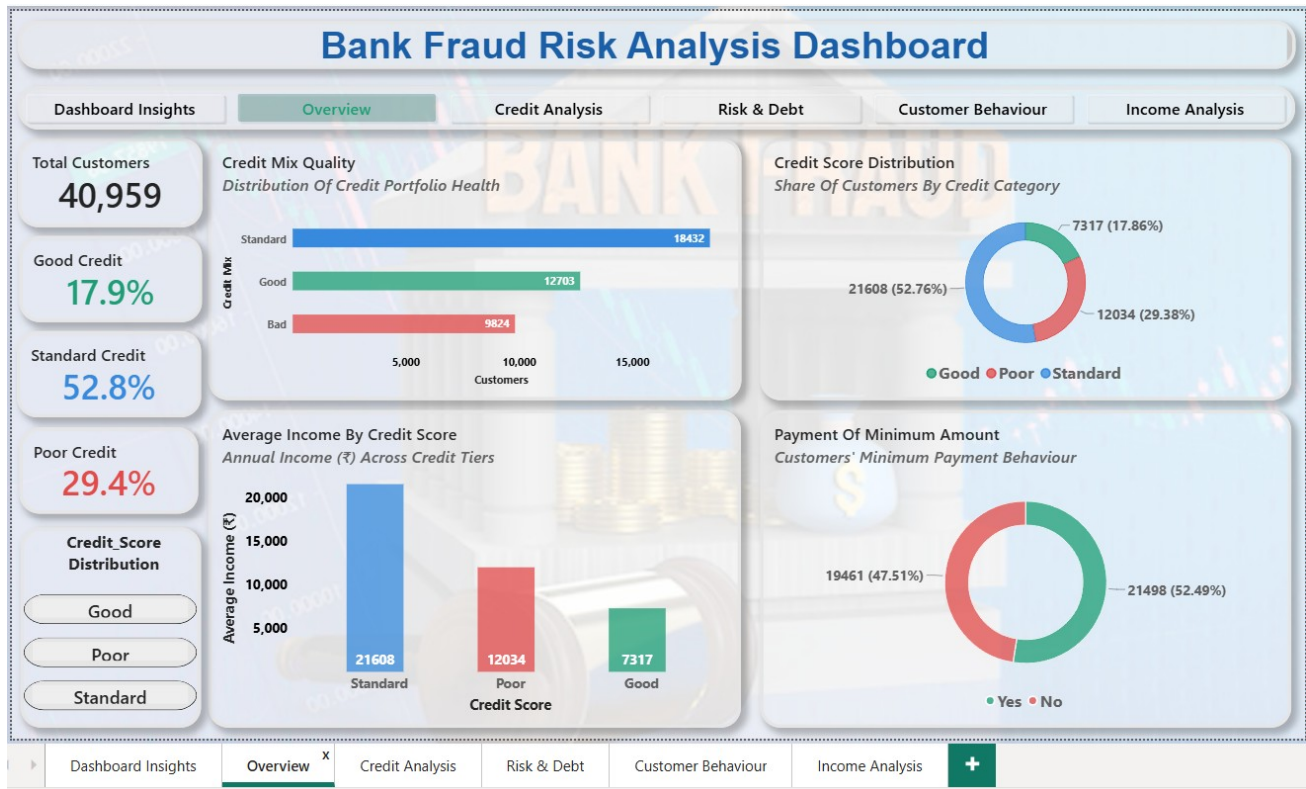
Page 1: Dashboard Insights

KPI cards (total customers, credit %), credit score donut, avg income bar, credit mix, min payment donut.



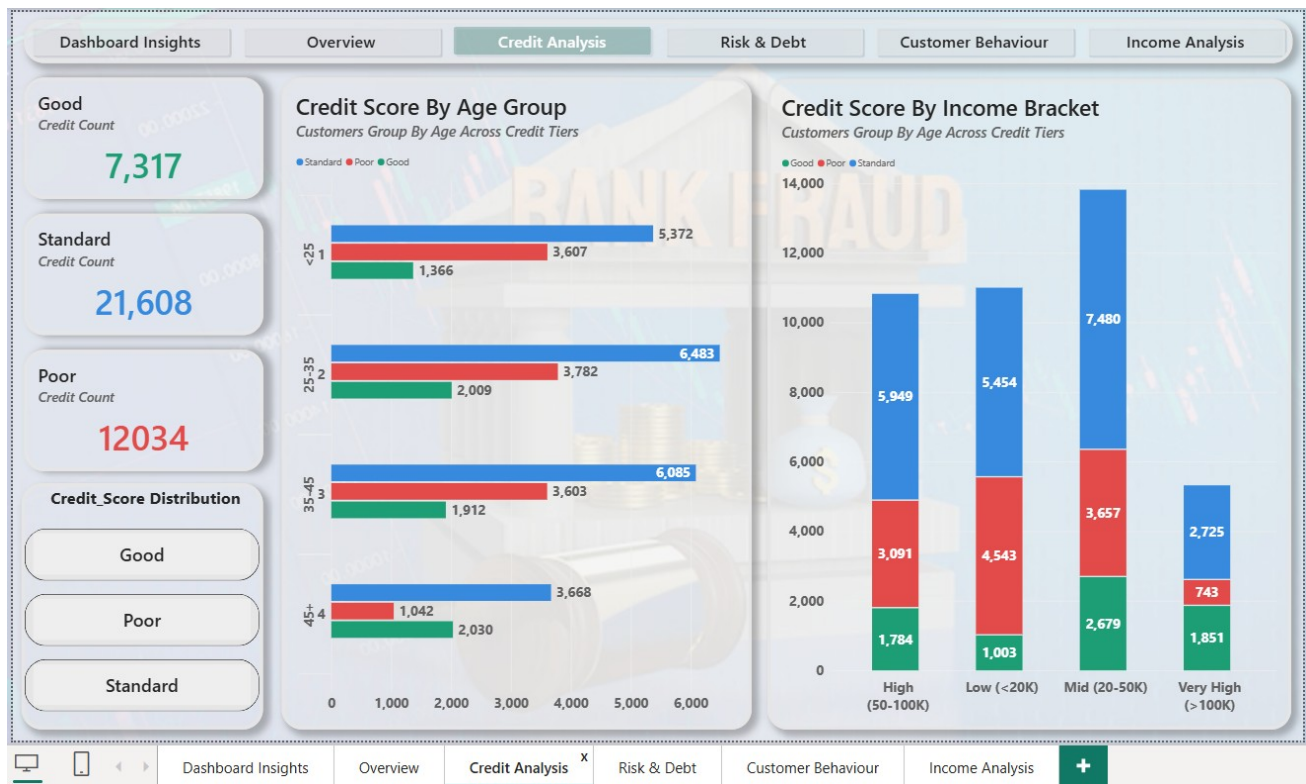
Page 2: Overview

KPI cards (total customers, credit %), credit score donut, avg income bar, credit mix, min payment donut.



Page 3: Credit Analysis

Stacked 100% bar by occupation, grouped bars by age group and income bracket.



Page 4: Risk & Debt

Debt, interest rate, delay days, EMI bars — all segmented by credit score. Insight text box.



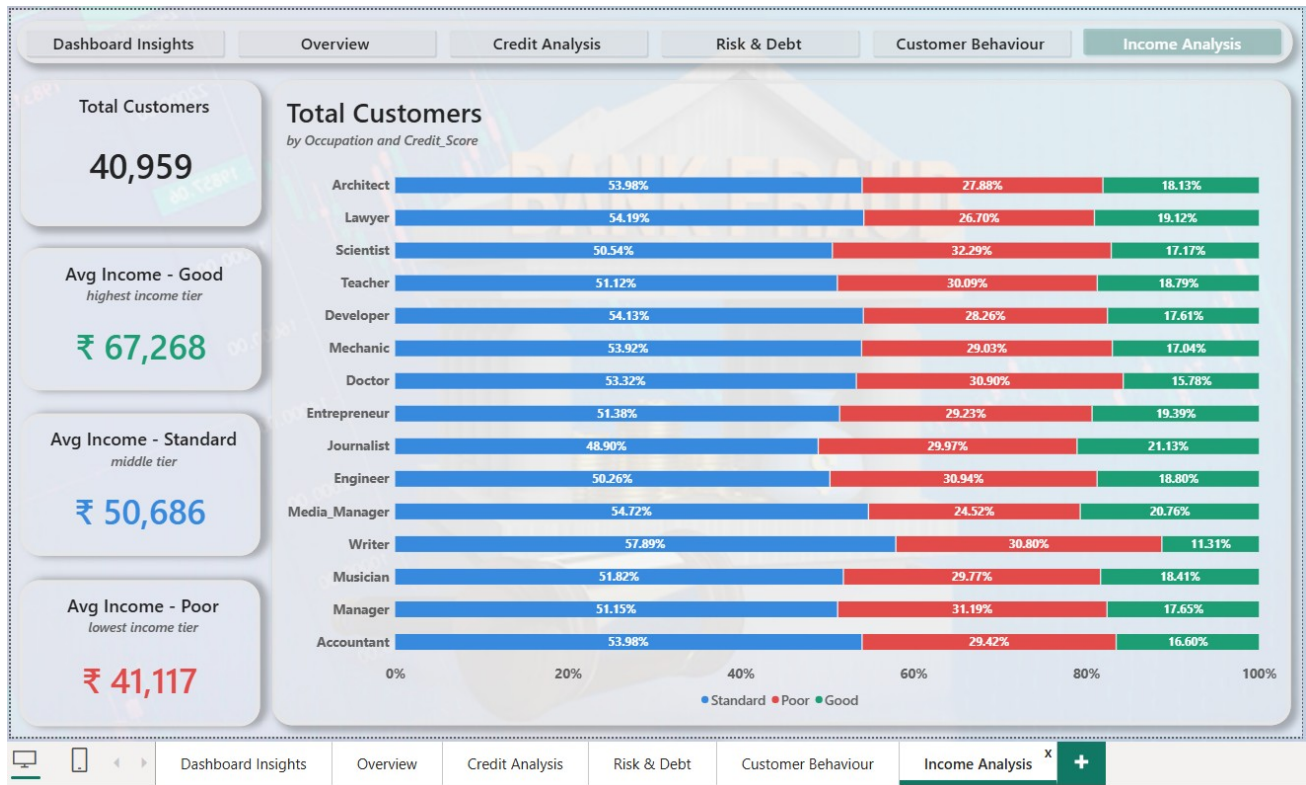
Page 5: Customer Behavior

Debt, interest rate, delay days, EMI bars — all segmented by credit score. Insight text box.



Page 6: Payment Behavior Analysis

Stacked 100% bar by occupation, grouped bars by age group and income bracket and sliced by the payment behavior of each category.



Step 4 — Key Insights & Explanations

4.1 Credit Score Distribution



Only 1 in 6 customers (17.9%) holds a Good credit score. The largest segment is Standard (52.8%), while nearly 1 in 3 customers (29.4%) falls in the high-risk Poor category. This distribution signals a significant portion of the portfolio that requires close monitoring for default and fraud risk.

4.2 Income vs. Credit Score

Finding: Income is a strong predictor of credit quality

Good credit holders earn an average of ₹67,268 per year — 63% more than Poor credit holders

Standard credit holders earn ₹50,683 — a middle position between the two extremes

Poor credit holders average only ₹41,117 — the lowest income tier

Recommendation: Flag loan applications from customers earning below ₹40,000 for enhanced due diligence

4.3 Risk & Debt Indicators

Metric	Good Credit	Standard Credit	Poor Credit	Risk Ratio
Avg Outstanding Debt	₹783	₹1,263	₹2,073	2.6× higher for Poor
Avg Interest Rate	7.58%	13.68%	19.96%	2.6× higher for Poor
Avg Payment Delay	10.9 days	19.8 days	29.6 days	2.7× higher for Poor
Avg Monthly EMI	₹125.6	₹101.2	₹111.5	Good pays highest EMI

Poor credit customers are caught in a debt trap cycle: high outstanding debt leads to high interest rates, which leads to payment delays, which further damages their credit score. These three metrics together are the most powerful fraud risk signals in this dataset.

4.4 Occupation Analysis

Credit score distribution is broadly similar across all 12 professions, indicating that occupation alone is not a decisive credit risk factor. However, some nuances exist:

- Journalists and Lawyers show a slightly higher proportion of Good credit (~18-19%)
- Scientists and Doctors show a slightly higher proportion of Poor credit (~23%)
- All occupations have Standard credit as the dominant category (50-55%)
- Recommendation: Use occupation as a secondary filter alongside income and debt metrics

4.5 Age Group Analysis

Finding: Younger customers carry disproportionately higher Poor credit rates

Customers under 25 have the highest Poor credit count (3,962) relative to their total group size

Customers aged 45+ have the lowest Poor credit count (705) — most financially stable group

This suggests age and financial maturity are positively correlated with credit quality

Recommendation: Apply stricter credit checks for applicants under 25 years of age

4.6 Payment Behavior Analysis

The 6 payment behavior types reveal customer spending and repayment personalities:

Payment Behaviour	Customers	Risk Profile
Low_spent_Small_value_payments	11,782 (28.8%)	Moderate risk — conservative spenders
High_spent_Medium_value_payments	8,101 (19.8%)	Watch — medium risk if credit is Poor
High_spent_Large_value_payments	5,920 (14.5%)	High risk if combined with Poor credit
Low_spent_Medium_value_payments	5,908 (14.4%)	Lower risk — controlled spending
High_spent_Small_value_payments	4,884 (11.9%)	Moderate risk
Low_spent_Large_value_payments	4,363 (10.7%)	Low risk — disciplined savers

35.5% of customers do NOT pay their minimum amount — this is the single strongest behavioural red flag for default and fraud risk. This metric should be used as a primary filter in any fraud scoring model.

Step 5 — Business Impact Summary

5.1 Strategic Value

Business Benefit	Estimated Impact
Faster credit decisions	Reduce approval time by 40-60%
Proactive fraud detection	Flag 29.4% high-risk portfolio segment early
Reduced default rate	Target collections on 35.5% non-min-payers
Personalized interest rate strategy	Rate range from 7.58% to 19.96% by risk tier
Age-based risk targeting	Tighter controls for <25 age group
Occupation-based loan product design	Customize sproducts for each profession segment

5.2 Dashboard KPI Summary

29.4% High-Risk Portfolio	2.6× Debt Ratio (Poor/Good)	35.5% Skip Min Payment	19.96% Max Avg Interest Rate
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5.3 Recommended Actions

- Implement an automated fraud alert for customers with delay > 25 days AND debt > ₹1,800
- Create a minimum payment monitoring workflow for the 35.5% non-payers
- Apply tiered interest rates formally aligned to the three credit tiers
- Develop a targeted financial wellness program for customers under 25
- Use this dashboard monthly as part of the credit committee review process
- Integrate the Power BI report with the loan origination system for real-time scoring

Executive Summary

40,959 customers analyzed across 5 dashboard pages in Power BI

29.4% of the portfolio (12,034 customers) are high-risk Poor credit holders

Poor credit customers carry 2.6× more debt and pay 2.7× more interest than Good credit holders

35.5% of customers skip minimum payments — the strongest fraud/default predictor in the dataset

This dashboard enables Bank to proactively manage credit risk, reduce defaults, and optimize their lending strategy across ₹40K+ customer records